CREATING LINEAR MODELS FROM WORD PROBLEMS

DOCUMENT 2 — FULL SOLUTIONS & EXPLANATIONS

Q1. Correct Answer: B | EASY

STEP-BY-STEP SOLUTION:

- Identify the flat (one-time) fee: $3.50 - this is the y-intercept (paid once).
- Identify the per-mile rate: $2.25 - this is the slope (multiplied by distance).
- Model: $C = 3.50 + 2.25m$.
- At $m=0$: C=$3.50 (just the flat fee). At $m=10$: $C=3.50+22.50=$26.00. Reasonable.

WHY EACH OPTION IS WRONG:

A) [WRONG] Swaps roles: $2.25 is added once, $3.50 is the per-mile charge -- both reversed.
B) [CORRECT] CORRECT. $3.50 is the one-time flat fee (y-intercept); $2.25 per mile is the slope.
C) [WRONG] Adds both values into one per-mile rate ($5.75/mi), losing the flat-fee structure.
D) [WRONG] Puts $3.50 as the per-mile rate and $2.25 as the flat fee -- roles reversed.

■ TRAP ALERT:

Students swap the flat fee and per-mile charge. Ask: 'Is this paid ONCE or EVERY mile?'

■ KEY INSIGHT:

The one-time charge is always the y-intercept (b). The per-unit rate is always the slope (m) in $y = mx + b$.

■ FAST METHOD:

Ask: Once = b. Every unit = m. Here $3.50 = once = b; $2.25 = every mile = m.

Q2. Correct Answer: C EASY

STEP-BY-STEP SOLUTION:

- The candle starts at 18 inches: initial value = 18.
- It burns DOWN (height decreases) at 0.4 in/hr: slope is NEGATIVE = -0.4.
- Model: $h = 18 - 0.4t$.
- Check: $t=0$ gives $h=18$ (full candle). $t=45$ gives $h=0$ (candle gone). Correct.

WHY EACH OPTION IS WRONG:

A) [WRONG] $h=0.4t+18$ has a POSITIVE slope, meaning the candle grows taller -- impossible.

B) [WRONG] Puts t in the wrong position; treats 18 as per-hour rate and 0.4 as a constant.

C) [CORRECT] CORRECT. Starts at 18 inches, decreases by 0.4 per hour.

D) [WRONG] $h=0.4-18t$ starts near zero with an enormous negative slope -- nonsensical.

■ TRAP ALERT:

Forgetting the negative sign when something burns, drains, or depletes. 'Burns DOWN' = negative slope.

■ KEY INSIGHT:

Decreasing quantities (burning, draining, depreciating) always have a NEGATIVE slope.

■ FAST METHOD:

Burning/draining/spending: subtract. Growing/filling/earning: add.

Q3. **Correct Answer: C** EASY

STEP-BY-STEP SOLUTION:

- Entry fee is a flat, one-time charge: \$4.00 = y-intercept.
- Per-interval charge: \$1.50 x h intervals = slope term.
- Total: 4.00 + 1.50h.

WHY EACH OPTION IS WRONG:

A) [WRONG] 4.00h + 1.50 uses entry fee as per-interval rate and \$1.50 as flat fee -- both swapped.

B) [WRONG] Ignores the entry fee entirely; only accounts for the per-interval charge.

C) [CORRECT] CORRECT. \$4.00 one-time entry fee + \$1.50 for each interval.

D) [WRONG] Adds both into one per-interval charge (\$5.50h) -- loses the flat-fee structure.

■ TRAP ALERT:

Combining both fees into one per-unit rate instead of keeping the flat fee as a separate constant.

■ KEY INSIGHT:

Flat/entry/registration fees always stand alone as the constant. Per-unit charges get multiplied by the variable.

■ FAST METHOD:

Flat fee = constant. Per-interval charge = coefficient of variable.

Q4. Correct Answer: 3 EASY

STEP-BY-STEP SOLUTION:

- Find the slope: m = (27 - 11) / (6 - 2) = 16 / 4 = 4.
- Use point-slope with (2, 11): 11 = 4(2) + b => b = 11 - 8 = 3.
- So f(x) = 4x + 3.
- Verify: f(6) = 4(6) + 3 = 27. Correct.
- f(0) = 4(0) + 3 = 3.

WHY EACH OPTION IS WRONG:

--) [WRONG] Error 1: Using one x-value (2 or 6) instead of the CHANGE in x (4) when computing slope.

--) [WRONG] Error 2: Assuming f(0) = 0. In a linear function, f(0) = b (the y-intercept), which need not be 0.

■ TRAP ALERT:

Students use a single x-value (like 6) as the denominator for slope instead of the change in x (6-2=4).

■ KEY INSIGHT:

f(0) = b = the y-intercept. Once you have slope m and any point, find b by substituting into y = mx + b.

■ FAST METHOD:

Slope = (27-11)/(6-2) = 4. Then b = 11 - 4(2) = 3. So f(0) = 3.

Q5. Correct Answer: B EASY

STEP-BY-STEP SOLUTION:

- Set total costs equal: 40 + 25m = 10 + 35m.
- Subtract 25m: 40 = 10 + 10m.
- Subtract 10: 30 = 10m => m = 3.
- Verify: Gym A = 40+75=$115. Gym B = 10+105=$115. Equal.

WHY EACH OPTION IS WRONG:

A) [WRONG] m=2: Gym A=$90, Gym B=$80. Not equal.

B) [CORRECT] CORRECT. After 3 months both cost $115.

C) [WRONG] m=4: Gym A=$140, Gym B=$150. Not equal.

D) [WRONG] m=5: Gym A=$165, Gym B=$185. Not equal.

■ TRAP ALERT:

Setting up 40 + 25 = 10 + 35 (forgetting to multiply by m) -- this gives m=0.

■ KEY INSIGHT:

Set the two full cost expressions equal and solve for m. Always verify by plugging back in.

■ FAST METHOD:

Difference in join fees: 30. Difference in monthly rates: 10. m = 30/10 = 3.

Q6. **Correct Answer: C** EASY

STEP-BY-STEP SOLUTION:

- ■ Starts with 240 rolls: initial value = 240.
- ■ Sells (removes) 15 per hour: slope = -15.
- ■ R = 240 - 15h.

WHY EACH OPTION IS WRONG:

A) [WRONG] R=15h-240 is negative at h=1 (-225 rolls) -- physically impossible.

B) [WRONG] R=240+15h means inventory grows with sales -- contradicts the scenario.

C) [CORRECT] CORRECT. Starts at 240, decreases by 15 per hour.

D) [WRONG] R=15-240h has an enormous negative rate -- nonsensical.

■ TRAP ALERT:

Writing +15h instead of -15h. Selling means inventory DECREASES.

■ KEY INSIGHT:

Physical decrease (selling, consuming, burning) always means a NEGATIVE slope.

■ FAST METHOD:

Starting amount minus (rate x time) = 240 - 15h.

Q7. Correct Answer: C EASY

STEP-BY-STEP SOLUTION:

- Hourly rate \$12.50 multiplied by hours worked: 12.50h.
- One-time signing bonus \$75: constant term +75.
- E = 12.50h + 75.

WHY EACH OPTION IS WRONG:

A) [WRONG] E=75h+12.50 treats the bonus as a per-hour rate -- the bonus is received once, not every hour.

B) [WRONG] E=12.50+75h treats the hourly rate as a constant -- Maria doesn't earn \$75 every hour.

C) [CORRECT] CORRECT. \$12.50 x hours worked plus the one-time \$75 bonus.

D) [WRONG] Combines both into one rate (\$87.50/hr) -- incorrect.

TRAP ALERT:

Adding both values into a single per-hour rate: 12.50 + 75 = 87.50h.

KEY INSIGHT:

A one-time payment (bonus, fee) is always the constant term (b). A recurring per-unit payment is always the slope (m).

FAST METHOD:

Label each amount: \$12.50/hr = m. \$75 one-time = b. Write E = 12.50h + 75.

Q8. Correct Answer: D EASY

STEP-BY-STEP SOLUTION:

- From 8:00 AM to 1:00 PM = 5 hours elapsed.
- Temperature rises 3 degrees per hour for 5 hours: 3 x 5 = 15 degrees increase.
- Temperature at 1 PM: 62 + 15 = 77 degrees F.

WHY EACH OPTION IS WRONG:

A) [WRONG] 65 degrees: only 1 hour counted (8 to 9 AM).

B) [WRONG] 68 degrees: only 2 hours counted.

C) [WRONG] 72 degrees: 10/3 hours -- not a whole number from 8 AM to 1 PM.

D) [CORRECT] CORRECT. 5 hours x 3 deg/hr = 15; 62 + 15 = 77 degrees F.

■ TRAP ALERT:

Miscounting hours: 8, 9, 10, 11, 12, 1 = six numbers but only FIVE gaps (hours). End minus start = 1PM - 8AM = 5 hours.

■ KEY INSIGHT:

Elapsed hours = end time - start time (not counting of clock numbers). 1PM - 8AM = 5 hours.

■ FAST METHOD:

1 PM - 8 AM = 5 hrs. 62 + 5(3) = 77 degrees F.

Q9. Correct Answer: D EASY

STEP-BY-STEP SOLUTION:

- $2.50 per cookie sold (variable): 2.50c.
- $100 donation (one-time, fixed): constant term +100.
- F = 2.50c + 100.

WHY EACH OPTION IS WRONG:

A) [WRONG] F=100c+2.50 multiplies the donation per cookie -- the donation is received once, not per sale.

B) [WRONG] F=2.50c-100 subtracts the donation, treating it as a cost rather than revenue.

C) [WRONG] F=2.50+100c would give $102.50 for 1 cookie and $1,002.50 for 10 -- treats donation as per-cookie.

D) [CORRECT] CORRECT. Revenue per cookie x quantity + one-time donation.

TRAP ALERT:

Putting the donation as the coefficient of c (Choice C), as if the donation were received for each cookie sold.

KEY INSIGHT:

A fixed donation is a constant added to the model regardless of how many units are sold.

FAST METHOD:

Sales revenue + fixed donation = 2.50c + 100.

Q10. Correct Answer: 50 EASY

STEP-BY-STEP SOLUTION:

- ■ Set up equation: 500 - 8t = 100.
- ■ 500 - 100 = 8t => 400 = 8t.
- ■ t = 400 / 8 = 50 minutes.

WHY EACH OPTION IS WRONG:

--) [WRONG] Error: Computing 500/8 = 62.5 instead of (500-100)/8 = 50.

--) [WRONG] Error: Subtracting 8 from 500 repeatedly rather than solving algebraically.

■ TRAP ALERT:

Dividing the full 500 gallons by 8 instead of first finding the amount to remove (500-100=400).

■ KEY INSIGHT:

Set the model equal to the target value, then solve. Tank starts at 500, must reach 100: remove 400 gallons.

■ FAST METHOD:

Need to remove 500-100=400 gallons at 8 gal/min. 400 / 8 = 50 minutes.

Q11. Correct Answer: B MEDIUM

STEP-BY-STEP SOLUTION:

- Standard total after n months: 9.99n.
- Promotional total after n months: 14.99 + 7.49n.
- Promo < Standard: 14.99 + 7.49n < 9.99n => 14.99 < 2.50n.
- n > 5.996. First full month where promo is less: n = 6.
- Verify: n=5: Standard=$49.95, Promo=$52.44 (Promo still MORE). n=6: Standard=$59.94, Promo=$59.93 (Promo cheaper by $0.01).

WHY EACH OPTION IS WRONG:

A) [WRONG] At n=5: Promo=$52.44 > Standard=$49.95. Promo is still MORE expensive.

B) [CORRECT] CORRECT. At n=6: Promo=$59.93 < Standard=$59.94. First month promo is cheaper.

C) [WRONG] n=7 is correct but is NOT the first month (n=6 already satisfies the condition).

D) [WRONG] n=8 also works but is not the first such month.

TRAP ALERT:

Solving n > 5.996 and rounding DOWN to 5. For 'first month condition is met,' always round UP.

KEY INSIGHT:

When solving 'first time a condition is met' with fractional result, round UP (ceiling) to the next whole number.

FAST METHOD:

14.99 / (9.99 - 7.49) = 14.99 / 2.50 = 5.996. First integer above this = 6.

Q12. Correct Answer: B MEDIUM

STEP-BY-STEP SOLUTION:

- ■ Revenue from n shirts: 20n.
- ■ Total cost: production cost + setup = 8n + 120.
- ■ Profit P = Revenue - Cost = 20n - (8n + 120) = 12n - 120.

WHY EACH OPTION IS WRONG:

A) [WRONG] P=20n-120 ignores the $8 per-shirt production cost.

B) [CORRECT] CORRECT. Profit per shirt = $20-$8=$12. Minus $120 fixed setup. P=12n-120.

C) [WRONG] P=12n+120 adds setup cost as profit instead of subtracting it.

D) [WRONG] P=8n-120 uses only production cost as the profit rate, ignoring the $20 selling price.

■ TRAP ALERT:

Using revenue minus only the fixed cost, forgetting the per-unit production cost. Profit = Revenue - ALL costs.

■ KEY INSIGHT:

Profit = Revenue - Total Cost. Total Cost includes BOTH fixed (one-time) and variable (per-unit x quantity) costs.

■ FAST METHOD:

Profit per shirt = 20 - 8 = $12. P = 12n - 120.

Q13. Correct Answer: A MEDIUM

STEP-BY-STEP SOLUTION:

- Rate of depreciation per year: (28,000 - 20,500) / 3 = 7,500 / 3 = $2,500/year.
- Model: V = 28,000 - 2,500t, where t = years since purchase.
- At t = 7: V = 28,000 - 2,500(7) = 28,000 - 17,500 = $10,500.

WHY EACH OPTION IS WRONG:

A) [CORRECT] CORRECT. $10,500 after 7 years.

B) [WRONG] $12,000 corresponds to t = 6.4 years -- not an integer.

C) [WRONG] $13,000 corresponds to t = 6 years, not 7.

D) [WRONG] $15,500 corresponds to t = 5 years, not 7.

TRAP ALERT:

Using the $7,500 total drop over 3 years as the annual rate instead of dividing by 3 to get $2,500/year.

KEY INSIGHT:

Rate of change = delta(y) / delta(x). Here: change in value / change in years = 7,500 / 3 = $2,500/yr.

FAST METHOD:

Slope = (20,500-28,000)/(3-0) = -2,500/yr. At 7 yrs: 28,000 - 17,500 = $10,500.

Q14. Correct Answer: C MEDIUM

STEP-BY-STEP SOLUTION:

- Tank A volume after t minutes: 12 + 3.5t.
- Tank B volume after t minutes: 47 - 2.5t.
- Set equal: 12 + 3.5t = 47 - 2.5t.
- 6t = 35 => t = 35/6 = 5.833 minutes.
- Nearest answer choice: 6.

WHY EACH OPTION IS WRONG:

A) [WRONG] t=5: Tank A=29.5L, Tank B=34.5L -- not equal.

B) [WRONG] t=5.5: Tank A=31.25L, Tank B=33.25L -- not equal.

C) [CORRECT] CORRECT. Exact answer is 35/6 = 5.83 min; nearest answer option is 6.

D) [WRONG] t=7: Tank A=36.5L, Tank B=29.5L -- long past the crossing point.

■ TRAP ALERT:

Adding rates instead of summing them: writing 3.5 - 2.5 = 1 instead of 3.5 + 2.5 = 6 for the combined approach rate.

■ KEY INSIGHT:

When one quantity increases and another decreases, the rate at which they approach each other is the SUM of the individual rates.

■ FAST METHOD:

12 + 3.5t = 47 - 2.5t => 6t = 35 => t = 5.83. Nearest option = 6.

Q15. Correct Answer: B MEDIUM

STEP-BY-STEP SOLUTION:

- Total kWh needed for 90 miles at 4 mi/kWh: 90 / 4 = 22.5 kWh.
- Car starts with 0 charge. Must charge: 22.5 kWh.
- Charging rate: 2.5 kWh/min. Time: 22.5 / 2.5 = 9 minutes.
- Verify: 9 min x 2.5 kWh/min = 22.5 kWh => 22.5 x 4 = 90 miles. Exact.

WHY EACH OPTION IS WRONG:

A) [WRONG] 8 min gives 8 x 2.5 = 20 kWh => only 80 miles. Insufficient.

B) [CORRECT] CORRECT. 9 min gives 9 x 2.5 = 22.5 kWh => 22.5 x 4 = 90 miles exactly.

C) [WRONG] 10 min works but is NOT the minimum.

D) [WRONG] 11 min works but exceeds the minimum.

TRAP ALERT:

The question asks for the MINIMUM integer, so once the equation is solved exactly, confirm it meets (not just exceeds) the requirement.

KEY INSIGHT:

When a result divides evenly into an integer (22.5/2.5 = 9 exactly), that integer is the minimum -- no rounding needed.

FAST METHOD:

Need 22.5 kWh. Charge rate: 2.5 kWh/min. 22.5 / 2.5 = 9 minutes exactly.

Q16. Correct Answer: 19 MEDIUM

STEP-BY-STEP SOLUTION:

- ■ Substitute x = 4, f(x) = 7 into f(x) = -3x + b.
- ■ 7 = -3(4) + b.
- ■ 7 = -12 + b.
- ■ b = 7 + 12 = 19.

WHY EACH OPTION IS WRONG:

--) [WRONG] Error: Computing -3(4) = -12 correctly but then writing b = 7 - 12 = -5 (subtracting instead of adding).

--) [WRONG] Error: Ignoring the negative sign, computing 3(4)=12, then b=7-12=-5.

■ TRAP ALERT:

Sign error: -3(4) = -12, so b = 7 - (-12) = 7 + 12 = 19, NOT 7 - 12 = -5.

■ KEY INSIGHT:

When solving y = mx + b for b: b = y - mx. Here b = 7 - (-3)(4) = 7 + 12 = 19.

■ FAST METHOD:

b = y - mx = 7 - (-3)(4) = 7 + 12 = 19.

Q17. Correct Answer: B MEDIUM

STEP-BY-STEP SOLUTION:

- ■ Set charges equal: 85 + 55h = 65 + 70h.
- ■ 85 - 65 = 70h - 55h => 20 = 15h => h = 4/3 = 1.33...
- ■ The exact crossover is not an integer. Check integer values:
- ■ h=1: Plumber=$140, Electrician=$135. Plumber still MORE expensive.
- ■ h=2: Plumber=$195, Electrician=$205. Plumber is LESS expensive.
- ■ First integer where plumber charges less: h = 2.

WHY EACH OPTION IS WRONG:

A) [WRONG] h=1: Plumber=$140 > Electrician=$135. Plumber is more expensive here.

B) [CORRECT] CORRECT. h=2: Plumber=$195 < Electrician=$205. First integer where plumber is cheaper.

C) [WRONG] h=3 also has plumber cheaper, but h=2 is the first such integer.

D) [WRONG] h=4 works but is later.

■ TRAP ALERT:

Expecting a whole-number answer and not checking integer values on either side of the fractional crossover (4/3 = 1.33).

■ KEY INSIGHT:

When the exact crossover is not an integer, check integer values on each side to find the first one satisfying the condition.

■ FAST METHOD:

85+55h=65+70h => 15h=20 => h=4/3. Not an integer. First integer above 4/3 is 2. Check: Plumber=$195 < $205.

Q18. Correct Answer: B MEDIUM

STEP-BY-STEP SOLUTION:

- ■ Both trains travel TOWARD each other, so their speeds add: 75 + 60 = 135 mph combined.
- ■ Total distance to close together: 405 miles.
- ■ Time to meet: 405 / 135 = 3 hours.

WHY EACH OPTION IS WRONG:

A) [WRONG] 2.5 hrs: 2.5 x 135 = 337.5 miles covered -- they haven't met yet.

B) [CORRECT] CORRECT. 3 hrs x 135 mph = 405 miles. Trains meet exactly.

C) [WRONG] 3.5 hrs: 3.5 x 135 = 472.5 miles -- they would have already passed.

D) [WRONG] 4 hrs: 4 x 135 = 540 miles -- well past meeting point.

■ TRAP ALERT:

Using only one train's speed (405/75 = 5.4 or 405/60 = 6.75) instead of adding both speeds.

■ KEY INSIGHT:

When two objects move toward each other, their closing speed is the SUM of individual speeds.

■ FAST METHOD:

Combined speed = 75 + 60 = 135 mph. Time = 405 / 135 = 3 hours.

Q19. Correct Answer: C MEDIUM

STEP-BY-STEP SOLUTION:

- ■ For profit, require R > C: 14w > 6w + 450.
- ■ 8w > 450 => w > 56.25.
- ■ Minimum integer: w = 57.
- ■ Verify: w=56: R=784, C=786. Revenue < Cost (no profit).
- ■ w=57: R=798, C=792. R > C. Profit = $6. First profitable quantity.

WHY EACH OPTION IS WRONG:

A) [WRONG] w=55: R=770, C=780. Revenue still less than cost.

B) [WRONG] w=56: R=784, C=786. Revenue is $2 SHORT of cost -- no profit.

C) [CORRECT] CORRECT. w=57: R=798, C=792. First time revenue exceeds cost.

D) [WRONG] w=58 also works but is not the minimum.

■ TRAP ALERT:

Solving 8w > 450 => w > 56.25 and rounding DOWN to 56. At w=56, revenue is still less than cost. Must round UP.

■ KEY INSIGHT:

For strict inequalities ('revenue GREATER THAN cost'), round UP the fractional solution to next whole number.

■ FAST METHOD:

8w > 450 => w > 56.25 => minimum integer = 57.

Q20. Correct Answer: B MEDIUM

STEP-BY-STEP SOLUTION:

- Set P > 80: 2.4t + 38 > 80.
- 2.4t > 42 => t > 17.5.
- First integer t: t = 18.
- Year: 2010 + 18 = 2028.
- Verify: t=17: P=2.4(17)+38=78.8 < 80. t=18: P=2.4(18)+38=81.2 > 80.

WHY EACH OPTION IS WRONG:

A) [WRONG] t=17: P=78.8 thousand. Not yet exceeded 80.

B) [CORRECT] CORRECT. t=18: P=81.2 thousand. Year = 2010+18 = 2028.

C) [WRONG] 2029 (t=19): Population exceeded 80,000 a year earlier in 2028.

D) [WRONG] 2030: Two years after the actual first exceedance.

■ TRAP ALERT:

Forgetting to add t back to the base year 2010. The variable t = years SINCE 2010.

■ KEY INSIGHT:

Always convert t back to the actual year. If t=0 is 2010, then year = 2010 + t.

■ FAST METHOD:

2.4t > 42 => t > 17.5 => t=18 => year = 2010+18 = 2028.

Q21. Correct Answer: 200 MEDIUM

STEP-BY-STEP SOLUTION:

- Set costs equal: $30 + 0.10t = 0.25t$.
- $30 = 0.25t - 0.10t = 0.15t$.
- $t = 30 / 0.15 = 200$ texts.

WHY EACH OPTION IS WRONG:

--) [WRONG] Error: Subtracting rates in wrong order: $0.10 - 0.25 = -0.15$, giving negative t.

--) [WRONG] Error: Solving $30 = 0.35t$ (adding rates instead of subtracting) gives $t = 85.7$.

TRAP ALERT:

Subtracting rates incorrectly. Move ALL variable terms to one side: $30 = 0.25t - 0.10t = 0.15t$.

KEY INSIGHT:

When equating two linear functions, collect variable terms on one side: $30 = (0.25-0.10)t = 0.15t$.

FAST METHOD:

$30 = (0.25-0.10)t = 0.15t \Rightarrow t = 200$.

Q22. Correct Answer: B MEDIUM

STEP-BY-STEP SOLUTION:

- ■ Crew 1 total after h hours: 40 + 12h.
- ■ Crew 2 total after h hours: 18h.
- ■ Set Crew 2 > Crew 1: 18h > 40 + 12h.
- ■ 6h > 40 => h > 6.67.
- ■ First integer h where Crew 2 is ahead: h = 7.
- ■ Verify: h=6: Crew1=112 ft, Crew2=108 ft (Crew1 ahead). h=7: Crew1=124 ft, Crew2=126 ft (Crew2 ahead).

WHY EACH OPTION IS WRONG:

A) [WRONG] h=6: Crew1=112, Crew2=108. Crew2 is still BEHIND.

B) [CORRECT] CORRECT. h=7: Crew1=124, Crew2=126. First time Crew2 is ahead.

C) [WRONG] h=8 works but is not the first hour.

D) [WRONG] h=9 also works but is later.

■ TRAP ALERT:

Rounding 6.67 DOWN to 6 and incorrectly concluding Crew2 is ahead at h=6.

■ KEY INSIGHT:

When asked for 'first time' with a fractional crossover, round UP and verify with actual numbers.

■ FAST METHOD:

6h > 40 => h > 6.67 => minimum integer = 7.

Q23. Correct Answer: C MEDIUM

STEP-BY-STEP SOLUTION:

- In the model $m = 500 - 35t$:
- When $t = 0$ (administration time): $m = 500 - 35(0) = 500$ mg.
- So 500 is the initial amount of medication at the moment of administration.
- The slope -35 represents the rate of elimination (35 mg/hr).

WHY EACH OPTION IS WRONG:

A) [WRONG] -35 is the rate of elimination per hour, not 500.

B) [WRONG] Hours of effectiveness: set m=0 => t=500/35=14.3 hrs. That is not 500.

C) [CORRECT] CORRECT. At t=0, m=500 mg. This is the initial dose administered.

D) [WRONG] At t=35: m=500-35(35)=-725, which is negative and not meaningful.

TRAP ALERT:

Confusing the y-intercept (initial value) with the slope (rate). Always plug in t=0 to find the initial value.

KEY INSIGHT:

In a linear model y=mx+b, the y-intercept b is the value when x=0: the STARTING value.

FAST METHOD:

Plug in t=0: m=500-35(0)=500. This is the initial (starting) amount.

Q24. Correct Answer: B MEDIUM

STEP-BY-STEP SOLUTION:

- ■ Set Sasha > Devon: 110 + 80w > 350 + 45w.
- ■ 35w > 240 => w > 6.857.
- ■ First integer: w = 7.
- ■ Verify: w=6: Devon=620, Sasha=590 (Devon ahead). w=7: Devon=665, Sasha=670 (Sasha first ahead).

WHY EACH OPTION IS WRONG:

A) [WRONG] w=6: Devon=$620, Sasha=$590. Devon still ahead.

B) [CORRECT] CORRECT. w=7: Devon=$665, Sasha=$670. Sasha first exceeds Devon.

C) [WRONG] w=8 works but is not the least such value.

D) [WRONG] w=9 also works but is later.

■ TRAP ALERT:

240 / 35 = 6.857 -- rounding DOWN to 6 gives a week where Devon is still ahead. Must round UP.

■ KEY INSIGHT:

To find 'first time A exceeds B,' solve the inequality and take the ceiling (round up) of the fractional answer.

■ FAST METHOD:

35w > 240 => w > 6.857 => minimum w = 7.

Q25. Correct Answer: C MEDIUM

STEP-BY-STEP SOLUTION:

- ■ Net drain rate: 80 - 15 = 65 gallons per minute.
- ■ Volume to remove: 12,000 - 8,200 = 3,800 gallons.
- ■ Time: 3,800 / 65 = 58.46 minutes.
- ■ Nearest answer choice: 58.5 (Choice C).

WHY EACH OPTION IS WRONG:

A) [WRONG] 47.5 min: 47.5 x 65 = 3,087.5 gal removed => pool at 8,912 gal, not 8,200.

B) [WRONG] 56.9 min: 56.9 x 65 = 3,698.5 gal removed => pool at 8,302 gal, not 8,200.

C) [CORRECT] CORRECT. 58.5 x 65 = 3,802.5 => pool at ~8,197 gal, closest to 8,200.

D) [WRONG] 60 min: 60 x 65 = 3,900 => pool at 8,100 gal, overshoots the target.

■ TRAP ALERT:

Using only the 80 gal/min drain rate without subtracting the 15 gal/min inflow from the hose.

■ KEY INSIGHT:

Net rate = outflow rate - inflow rate. Always account for both directions when rates work in opposition.

■ FAST METHOD:

Net drain: 65 gal/min. Need to remove 3,800 gal. 3,800/65 = 58.46 => answer C (58.5).

Q26. Correct Answer: C HARD

STEP-BY-STEP SOLUTION:

- Set equal: $0.42m = 38 + 0.18m$.
- $0.24m = 38 \Rightarrow m = 38 / 0.24 = 158.33$ miles.
- At $m < 158.33$ (e.g., $m=100$): Plan A = $42, Plan B = $56. Plan A is cheaper.
- At $m > 158.33$ (e.g., $m=200$): Plan A = $84, Plan B = $74. Plan B is cheaper.
- So for FEWER miles than 158.33, Plan A is cheaper.

WHY EACH OPTION IS WRONG:

A) [WRONG] 152 miles is incorrect. Break-even is 38/0.24 = 158.33, not 152.

B) [WRONG] 158 miles is close but not exact (158.33). Also wrongly states Plan B cheaper at low mileage.

C) [CORRECT] CORRECT. 158.33 miles exact. For fewer miles, Plan A (no flat fee) is cheaper.

D) [WRONG] Correctly states 158.33 miles but wrongly claims Plan B is cheaper for fewer miles.

TRAP ALERT:

Confusing which plan is cheaper at low vs. high mileage. The plan with a flat fee is cheaper only at HIGH mileage.

KEY INSIGHT:

A plan with flat fee + lower per-unit rate is cheaper for LARGE quantities. No flat fee + higher per-unit rate is cheaper for SMALL quantities.

FAST METHOD:

Break-even: $0.42m = 38+0.18m \Rightarrow 0.24m=38 \Rightarrow m=158.33$. At $m<158.33$: Plan A costs less.

Q27. Correct Answer: B HARD

STEP-BY-STEP SOLUTION:

- Set Co.2 > Co.1: 12m + 90 > 8.5m + 124.
- 3.5m > 34 => m > 9.71.
- First integer: m = 10.
- Verify: m=9: Co.1=200.5, Co.2=198 (Co.1 higher). m=10: Co.1=209, Co.2=210 (Co.2 higher).

WHY EACH OPTION IS WRONG:

A) [WRONG] m=9: Co.2=$198K < Co.1=$200.5K. Not yet exceeded.

B) [CORRECT] CORRECT. m=10: Co.2=$210K > Co.1=$209K. First month Co.2 exceeds Co.1.

C) [WRONG] m=11 works but is not the first such month.

D) [WRONG] m=12 also works but is later.

■ TRAP ALERT:

Rounding 9.71 DOWN to 9 and not checking if Co.2 actually exceeds Co.1 at m=9.

■ KEY INSIGHT:

Always verify by plugging in both the value below (9) and the ceiling (10) to confirm the crossover.

■ FAST METHOD:

3.5m > 34 => m > 9.71 => m = 10.

Q28. Correct Answer: A HARD

STEP-BY-STEP SOLUTION:

- Total catering cost for 12 people: 18 x 12 = $216.
- Set up inequality: 75h + 216 <= 600.
- 75h <= 384 => h <= 5.12.
- Maximum whole hours: h = 5.
- Verify: h=5: 75(5)+216=591 <= 600. h=6: 75(6)+216=666 > 600.

WHY EACH OPTION IS WRONG:

A) [CORRECT] CORRECT. 75h + 216 <= 600, max h = 5.

B) [WRONG] Uses catering cost of $18 (one person) instead of $18x12=$216 -- ignores group size.

C) [WRONG] Sets up the inequality correctly (75h+216<=600) but miscalculates: h<=5.12 rounds to 5, not 4.

D) [WRONG] Adds $75+$18=$93 per hour -- incorrectly treats catering as a per-hour rate.

TRAP ALERT:

Forgetting to multiply the per-person catering cost by the number of attendees: 18 x 12 = $216, not just $18.

KEY INSIGHT:

When a per-unit cost applies to multiple items (people, units), always multiply first to get the total for that component.

FAST METHOD:

Catering total = 18x12 = $216. 75h <= 600-216 = 384. h <= 5.12 => max = 5 hours.

Q29. Correct Answer: C HARD

STEP-BY-STEP SOLUTION:

- Slope: m = (52 - 17) / (9 - 2) = 35 / 7 = 5.
- Use f(2)=17: 17 = 5(2) + b => 17 = 10 + b => b = 7.
- f(x) = 5x + 7.
- f(0) = 5(0) + 7 = 7.

WHY EACH OPTION IS WRONG:

A) [WRONG] b=3 results from using wrong denominator: 35/9 instead of 35/7.

B) [WRONG] b=5 confuses the slope value with the y-intercept value.

C) [CORRECT] CORRECT. m=5, b=7, so f(0)=7.

D) [WRONG] b=9 comes from a sign error in computing b.

■ TRAP ALERT:

Using 9 (an x-value) instead of 9-2=7 (the CHANGE in x) when computing slope.

■ KEY INSIGHT:

Slope = (change in y) / (change in x) = (y2-y1)/(x2-x1). Always use both differences.

■ FAST METHOD:

m = (52-17)/(9-2) = 35/7 = 5. b = 17-5(2) = 7. f(0) = 7.

Q30. Correct Answer: C HARD

STEP-BY-STEP SOLUTION:

- Newsletter 2 started at t=4. At calendar time t, its elapsed time is (t-4).
- Newsletter 2 subscribers: S2 = 180(t - 4) for t >= 4.
- Set equal: 120t + 80 = 180(t - 4).
- 120t + 80 = 180t - 720.
- 800 = 60t => t = 800/60 = 13.33.
- Check t=13: S1=1,640, S2=180(9)=1,620. S1 ahead. t=14: S1=1,760, S2=180(10)=1,800. S2 first exceeds S1.

WHY EACH OPTION IS WRONG:

A) [WRONG] t=10: S1=1,280, S2=180(6)=1,080. S2 still behind.

B) [WRONG] t=12: S1=1,520, S2=180(8)=1,440. S2 still behind.

C) [CORRECT] CORRECT. t=14: S2=1,800 > S1=1,760. First time S2 exceeds S1.

D) [WRONG] t=16: S2=2,160 > S1=2,000. Correct, but not the first such time.

TRAP ALERT:

Setting up newsletter 2's equation without accounting for its delayed start. Use (t-4), not t.

KEY INSIGHT:

When a second process starts later, its elapsed time at calendar time t is (t - start time).

FAST METHOD:

120t+80=180(t-4) => 800=60t => t=13.33. First integer check at t=14.

Q31. Correct Answer: 14 HARD

STEP-BY-STEP SOLUTION:

- Time = distance / speed. Set times equal: 72/(r+4) = 40/(r-4).
- Cross-multiply: 72(r-4) = 40(r+4).
- 72r - 288 = 40r + 160.
- 32r = 448 => r = 14 mph.
- Verify: 72/(14+4)=72/18=4 hrs. 40/(14-4)=40/10=4 hrs. Equal.

WHY EACH OPTION IS WRONG:

--) [WRONG] Error: Cross-multiplying as 72(r+4)=40(r-4) -- reversing which fraction is on which side.

--) [WRONG] Error: Adding distances and speeds rather than setting times equal.

TRAP ALERT:

Setting up 72(r+4) = 40(r-4) instead of 72(r-4) = 40(r+4) -- reversing the cross multiplication.

KEY INSIGHT:

When times are equal and time=dist/speed, set the two time fractions equal and cross-multiply carefully.

FAST METHOD:

72/(r+4) = 40/(r-4) => 72r-288 = 40r+160 => 32r=448 => r=14.

Q32. Correct Answer: A HARD

STEP-BY-STEP SOLUTION:

- ■ Break-even: Revenue = Cost => 22x = 14x + 3,200.
- ■ 8x = 3,200 => x = 400 units.
- ■ The constant 3,200 is the term independent of x (fixed/overhead cost: rent, equipment, etc.).
- ■ Revenue at break-even: 22(400) = $8,800 (not $3,200).

WHY EACH OPTION IS WRONG:

A) [CORRECT] CORRECT. 400 units; $3,200 is the fixed overhead cost.

B) [WRONG] 229 units: 22(229)=5,038 and cost=14(229)+3200=6,406. Not break-even.

C) [WRONG] Revenue at break-even = $8,800, not $3,200.

D) [WRONG] 145 units: 22(145)=3,190 and cost=14(145)+3200=5,230. Not break-even.

■ TRAP ALERT:

Dividing the fixed cost by selling price (3,200/22=145) instead of solving Revenue=Cost for x.

■ KEY INSIGHT:

Break-even: set Revenue = Cost and solve for x. The constant in the cost equation = fixed overhead cost.

■ FAST METHOD:

22x=14x+3,200 => 8x=3,200 => x=400. 3,200=fixed cost.

Q33. Correct Answer: D HARD

STEP-BY-STEP SOLUTION:

- Slope: m = (-14 - (-2)) / (7 - 3) = -12 / 4 = -3.
- Using (3, -2): -2 = -3(3) + b => -2 = -9 + b => b = 7.
- g(x) = -3x + 7.
- X-intercept: set g(x) = 0 => 0 = -3x + 7 => x = 7/3.
- The exact x-intercept is (7/3, 0) = (2.33..., 0).

WHY EACH OPTION IS WRONG:

A) [WRONG] (-1/2, 0): g(-1/2)=-3(-0.5)+7=1.5+7=8.5 not equal to 0.

B) [WRONG] (1/2, 0): g(0.5)=-3(0.5)+7=5.5 not equal to 0.

C) [WRONG] (1, 0): g(1)=-3(1)+7=4 not equal to 0.

D) [CORRECT] CORRECT. g(7/3)=-3(7/3)+7=-7+7=0. The x-intercept is exactly (7/3, 0).

TRAP ALERT:

Making sign errors when computing b: -2 = -3(3) + b => -2 = -9 + b => b = +7, not -7.

KEY INSIGHT:

To find x-intercept, set y=0 and solve for x. X-intercept is where the graph CROSSES the x-axis.

FAST METHOD:

m=-12/4=-3. b=-2-(-3)(3)=-2+9=7. Set 0=-3x+7 => x=7/3.

Q34. Correct Answer: B HARD

STEP-BY-STEP SOLUTION:

- ■ I = R - C = (20n - 200) - (500 + 12n) = 20n - 200 - 500 - 12n = 8n - 700.
- ■ Set I >= 1,000: 8n - 700 >= 1,000.
- ■ 8n >= 1,700 => n >= 212.5.
- ■ Minimum integer: n = 213.
- ■ Verify: n=212: I=8(212)-700=1,696-700=996 < 1,000. n=213: I=8(213)-700=1,004 >= 1,000.

WHY EACH OPTION IS WRONG:

A) [WRONG] n=212: I=$996 < $1,000. Does not meet the target.

B) [CORRECT] CORRECT. n=213: I=$1,004 >= $1,000. First quantity meeting the target.

C) [WRONG] n=214 works but is not the minimum.

D) [WRONG] n=215 also works but is not the minimum.

■ TRAP ALERT:

Forgetting that Net Income = Revenue MINUS Cost. Students sometimes add them or use only revenue.

■ KEY INSIGHT:

Net income (profit) = Revenue - Total Cost. Always subtract ALL costs from revenue.

■ FAST METHOD:

I=(20n-200)-(500+12n)=8n-700 >= 1,000 => n >= 212.5 => n=213.

Q35. Correct Answer: B HARD

STEP-BY-STEP SOLUTION:

- Energy generated: 0.3 kWh/hr x 9 hrs = 2.7 kWh.
- This represents 60% of daily usage D: 2.7 = 0.60 x D.
- D = 2.7 / 0.60 = 4.5 kWh.

WHY EACH OPTION IS WRONG:

A) [WRONG] 3.2 kWh: incorrect percentage setup or arithmetic error.

B) [CORRECT] CORRECT. 0.3x9=2.7 kWh = 60% of D => D=2.7/0.6=4.5 kWh.

C) [WRONG] 4.7: comes from rounding error or using 58% instead of 60%.

D) [WRONG] 5.4 kWh: results from dividing 2.7 by 0.5 (50%) instead of 0.6 (60%).

■ TRAP ALERT:

Using 60% as a multiplier of the answer (D x 0.6 = 2.7) but then forgetting to DIVIDE by 0.6 to isolate D.

■ KEY INSIGHT:

If A is P% of B, then B = A / (P/100). Here: daily usage = 2.7 / 0.6 = 4.5 kWh.

■ FAST METHOD:

0.3 x 9 = 2.7 kWh = 60% of D. D = 2.7/0.6 = 4.5 kWh.

Q36. Correct Answer: A HARD

STEP-BY-STEP SOLUTION:

- Slope of line l: m = (-1 - 7) / (5 - (-3)) = -8 / 8 = -1.
- Line k is parallel to l, so k also has slope = -1.
- Line k passes through (0, -4): k has equation y = -x - 4.
- Line l: using point (5, -1): -1 = -1(5) + b => b = 4. So l: y = -x + 4.
- Both lines have slope -1 but different y-intercepts (4 vs. -4). They are parallel => NO intersection.

WHY EACH OPTION IS WRONG:

A) [CORRECT] CORRECT. Same slope (-1), different y-intercepts. Parallel lines never intersect.

B) [WRONG] (1, -5): on k? y=-1-4=-5. On l? y=-1+4=3 not equal to -5. Not on both.

C) [WRONG] (-3, -1): l passes through (-3, 7), not (-3, -1).

D) [WRONG] (0, 4): this is l's y-intercept, but on k: y=0-4=-4, not 4.

TRAP ALERT:

Attempting to solve the system anyway and getting confused. Parallel lines HAVE NO solution.

KEY INSIGHT:

Parallel lines have equal slopes. If slopes are equal and y-intercepts differ, there is NO intersection.

FAST METHOD:

Both lines slope=-1 but y-intercepts 4 and -4 differ => parallel => no intersection.

Q37. Correct Answer: 4 HARD

STEP-BY-STEP SOLUTION:

- ■ Slope a = (-11 - 13) / (6 - (-2)) = -24 / 8 = -3.
- ■ Use p(6) = -11: -11 = -3(6) + b => -11 = -18 + b => b = 7.
- ■ Verify: p(-2) = -3(-2) + 7 = 6 + 7 = 13. Correct.
- ■ a + b = -3 + 7 = 4.

WHY EACH OPTION IS WRONG:

--) [WRONG] Error: Computing slope a=-3 and gridding that in directly, forgetting to add b.

--) [WRONG] Error: Sign mistake in computing b: -11=-18+b => b=-11+18=7 (correct), but some write b=-7.

■ TRAP ALERT:

Stopping after finding slope a=-3 and gridding -3 as the answer, without completing the step of finding b and computing a+b.

■ KEY INSIGHT:

Multi-step problems: complete EVERY step. Find a, then find b, THEN compute a+b. Here: -3+7=4.

■ FAST METHOD:

a=(-11-13)/(6-(-2))=-24/8=-3. b=-11-(-3)(6)=-11+18=7. a+b=-3+7=4.

Q38. Correct Answer: B HARD

STEP-BY-STEP SOLUTION:

- x = years since 2015. P is in millions.
- Slope: $m = (7.2 - 3.6) / (5 - 2) = 3.6 / 3 = 1.2$ million/year.
- Use $P(2)=3.6$: $3.6 = 1.2(2) + b \Rightarrow 3.6 = 2.4 + b \Rightarrow b = 1.2$.
- Model: $P = 1.2x + 1.2$.
- Set $P=0$: $0 = 1.2x + 1.2 \Rightarrow x = -1$.
- Year: $2015 + (-1) = 2014$.

WHY EACH OPTION IS WRONG:

A) [WRONG] 2013 ($x=-2$): $P=1.2(-2)+1.2=-2.4+1.2=-1.2$. Not zero.

B) [CORRECT] CORRECT. $x=-1 \Rightarrow$ year=2015-1=2014. $P=1.2(-1)+1.2=0$.

C) [WRONG] 2015 ($x=0$): $P(0)=1.2$. Not zero.

D) [WRONG] 2016 ($x=1$): $P=2.4$. Not zero.

TRAP ALERT:

Forgetting that a negative x means a year BEFORE the base year. $x=-1 \Rightarrow 2015-1=2014$, not 2016.

KEY INSIGHT:

When $t=0$ corresponds to a specific year, a negative t goes backward in time: year = base year + t.

FAST METHOD:

$m=3.6/3=1.2$. $b=3.6-1.2(2)=1.2$. Set $0=1.2x+1.2 \Rightarrow x=-1 \Rightarrow$ year=2014.

Q39. Correct Answer: 11 HARD

STEP-BY-STEP SOLUTION:

- ■ Rate of increase: (27 - 12) / (6 - 1) = 15 / 5 = 3 miles per week.
- ■ Model: D = 12 + 3(w - 1) = 3w + 9, where w is the week number.
- ■ Set D = 42: 42 = 3w + 9 => 3w = 33 => w = 11.

WHY EACH OPTION IS WRONG:

--) [WRONG] Error: slope = (27-12)/6 = 2.5, using 6 as denominator instead of 6-1=5.

--) [WRONG] Error: Model D=3w+12 (using week 0 as base) gives w=10 instead of 11.

■ TRAP ALERT:

Computing slope as (27-12)/6=2.5 rather than (27-12)/(6-1)=3. The weeks run from 1 to 6, a span of 5 weeks.

■ KEY INSIGHT:

When weeks/periods are numbered starting from 1, the span from week 1 to week 6 is 6-1=5 periods, not 6.

■ FAST METHOD:

m=15/5=3. D=3w+9. Set 42=3w+9 => w=11.

Q40. Correct Answer: D HARD

STEP-BY-STEP SOLUTION:

- Profit = Revenue - Cost = 7.50f - (3.60f + 280) = 3.90f - 280.
- Set profit >= 500: 3.90f - 280 >= 500.
- 3.90f >= 780 => f >= 200.
- At f=200: Profit = 3.90(200)-280 = 780-280 = $500. Exactly meets goal.
- Minimum integer: f = 200.

WHY EACH OPTION IS WRONG:

A) [WRONG] f=197: Profit=3.90(197)-280=488.3<500. Insufficient.

B) [WRONG] f=198: Profit=3.90(198)-280=492.2<500. Still insufficient.

C) [WRONG] f=199: Profit=3.90(199)-280=496.1<500. Still insufficient.

D) [CORRECT] CORRECT. f=200: Profit=3.90(200)-280=500. Exactly meets the savings goal.

TRAP ALERT:

Computing revenue (7.50f) minus only the fixed cost ($280), forgetting to subtract the variable cost ($3.60f).

KEY INSIGHT:

Profit per item = Selling price - Variable cost per item = $7.50 - $3.60 = $3.90. Use this in the profit equation.

FAST METHOD:

Profit=3.90f-280 >= 500 => 3.90f >= 780 => f >= 200. Minimum = 200.

Q41. Correct Answer: B HARD

STEP-BY-STEP SOLUTION:

- Rate of change from x=-4 to x=0: (10-22)/(0-(-4)) = -12/4 = -3.
- Confirm from x=0 to x=4: (-2-10)/(4-0) = -12/4 = -3. Consistent.
- h(0) = 10: at x=0 the function value is 10. In y=mx+b, f(0)=b = the y-intercept.
- So: rate = -3; h(0)=10 is the y-intercept.

WHY EACH OPTION IS WRONG:

A) [WRONG] Rate=-3 is correct, but h(0)=10 is the y-intercept, NOT the x-intercept. X-intercept is where h(x)=0.

B) [CORRECT] CORRECT. Rate=-3; h(0)=10 is the y-intercept (value when x=0).

C) [WRONG] Rate=+3 is wrong -- the function DECREASES as x increases.

D) [WRONG] Rate=-12 comes from reading only the change in y (-12) without dividing by the change in x (4).

■ TRAP ALERT:

Confusing x-intercept (where h(x)=0) and y-intercept (where x=0). h(0)=10 is the y-intercept.

■ KEY INSIGHT:

In any function, f(0) gives the y-intercept. To find the x-intercept, set f(x)=0 and solve for x.

■ FAST METHOD:

Rate=(10-22)/4=-3. h(0)=10 at x=0 => this is the y-intercept.

Q42. Correct Answer: A HARD

STEP-BY-STEP SOLUTION:

- Break-even first: 0.38m = 55 + 0.15m => 0.23m = 55 => m = 55/0.23 = 239.1 miles.
- Since 230 < 239.1 (below break-even), Plan X is cheaper at 230 miles.
- Plan X at 230 miles: 0.38 x 230 = $87.40.
- Plan Y at 230 miles: 55 + 0.15(230) = 55 + 34.50 = $89.50.
- Difference: $89.50 - $87.40 = $2.10. Plan X is cheaper by $2.10.

WHY EACH OPTION IS WRONG:

A) [CORRECT] CORRECT. Plan X=$87.40; Plan Y=$89.50. Plan X cheaper by $2.10.

B) [WRONG] Plan Y is NOT cheaper at 230 miles (below break-even).

C) [WRONG] $5.40 difference: incorrect arithmetic. 89.50-87.40=$2.10, not $5.40.

D) [WRONG] Plan Y is not cheaper here and $5.40 is the wrong difference.

TRAP ALERT:

Confusing which plan is cheaper below break-even. Below break-even, the no-flat-fee plan (X) is always cheaper.

KEY INSIGHT:

Always compute both costs at the given quantity, then compare directly. Don't rely on intuition.

FAST METHOD:

X=0.38(230)=$87.40. Y=55+0.15(230)=$89.50. Difference=$2.10. X is cheaper.

Q43. Correct Answer: C HARD

STEP-BY-STEP SOLUTION:

- Net outflow per month: 85,000 - 12,500 = $72,500.
- Amount to deplete: 4,200,000 - 3,000,000 = $1,200,000.
- Months needed: 1,200,000 / 72,500 = 16.55 months.
- After 16 months: 4,200,000 - 16(72,500) = 3,040,000. Still above $3M.
- After 17 months: 4,200,000 - 17(72,500) = 2,967,500 < 3,000,000.

WHY EACH OPTION IS WRONG:

A) [WRONG] After 16 months: fund = $3,040,000 > $3,000,000. Not yet below threshold.

B) [WRONG] Same as A -- 16 months is insufficient.

C) [CORRECT] CORRECT. After 17 months: $2,967,500 < $3,000,000. First month below threshold.

D) [WRONG] 18 months also works but is not the FIRST month below threshold.

■ TRAP ALERT:

Using only the $85,000 monthly withdrawal without adding back the $12,500 grant income.

■ KEY INSIGHT:

Net change per period = all outflows minus all inflows. Always account for both directions of cash flow.

■ FAST METHOD:

Net outflow: $72,500/mo. Need to lose $1,200,000. 1,200,000/72,500=16.55 => first complete month=17.

Q44. Correct Answer: C HARD

STEP-BY-STEP SOLUTION:

- In s = -2.5n + 95:
- The slope is -2.5.
- Slope = change in s per 1-unit increase in n.
- As n (UV hours) increases by 1, s changes by -2.5.
- Interpretation: each additional hour of UV exposure per week DECREASES skin health score by 2.5 points.

WHY EACH OPTION IS WRONG:

A) [WRONG] Slope is NEGATIVE (-2.5), not positive. Score DECREASES, not increases.

B) [WRONG] The -2.5 is the slope. At n=0, s=95 (y-intercept). -2.5 is not the y-intercept.

C) [CORRECT] CORRECT. Slope -2.5: for each additional UV hour, score drops by 2.5 points.

D) [WRONG] This confuses y-intercept (95) with an x-value. At n=0, s=95; 2.5 hours is unrelated to 95.

■ TRAP ALERT:

Confusing slope with y-intercept. The slope tells you the RATE of change per unit, not a starting value.

■ KEY INSIGHT:

In y=mx+b, m=slope=rate of change per unit of x. Negative slope means y DECREASES as x increases.

■ FAST METHOD:

Slope=-2.5: for each +1 UV hour, score changes by -2.5. Negative => score decreases.

Q45. Correct Answer: 0.1 HARD

STEP-BY-STEP SOLUTION:

- ■ k = slope = (T2 - T1) / (d2 - d1).
- ■ k = (92 - 62) / (450 - 150) = 30 / 300 = 0.1.
- ■ Interpretation: temperature increases by 0.1 degree F for every additional foot of depth.

WHY EACH OPTION IS WRONG:

--) [WRONG] Error: k = (92-62)/450 = 0.067 -- dividing by one d-value instead of the change in d.

--) [WRONG] Error: k = 30/150 = 0.2 -- using only the first depth value in the denominator.

■ TRAP ALERT:

Dividing the change in T by a single depth value (150 or 450) instead of the CHANGE in depth (450-150=300).

■ KEY INSIGHT:

Rate of change (slope) always uses DIFFERENCES: k = delta(T) / delta(d) = (T2-T1)/(d2-d1).

■ FAST METHOD:

k=(92-62)/(450-150)=30/300=0.1 degrees per foot.

Q46. Correct Answer: B HARD

STEP-BY-STEP SOLUTION:

- Net pallets lost per day: 14 - 6 = 8.
- Let p = starting pallets. After 18 days: p - 8(18) = 100.
- p - 144 = 100 => p = 244.

WHY EACH OPTION IS WRONG:

A) [WRONG] 234: uses net rate of 7 instead of 8 (arithmetic error in 14-6).

B) [CORRECT] CORRECT. p = 100 + 8(18) = 100 + 144 = 244.

C) [WRONG] 252: uses only the 14 outflow rate without subtracting the 6 received.

D) [WRONG] 262: incorrectly adds both rates together.

■ TRAP ALERT:

Using 14/day as the net drain without subtracting the 6 pallets received each day.

■ KEY INSIGHT:

Net change = outflow - inflow. Work backwards to find start: starting = ending + net lost.

■ FAST METHOD:

Net loss: 8/day. In 18 days: 8(18)=144 lost. Start: 100+144=244.

Q47. Correct Answer: C HARD

STEP-BY-STEP SOLUTION:

- Find original hours h from \$645 quote: 645=150+45h => 45h=495 => h=11 hours.
- Client adds 2 more hours: new total hours = 11 + 2 = 13.
- New total fee: F = 150 + 45(13) = 150 + 585 = \$735.
- Alternative: 645 + 2(45) = 645 + 90 = \$735.

WHY EACH OPTION IS WRONG:

A) [WRONG] \$690 = \$645 + \$45. This adds only ONE extra hour, not two.

B) [WRONG] \$725 = \$645 + \$80. Uses incorrect additional rate.

C) [CORRECT] CORRECT. Two more hours x \$45/hr = \$90 extra. \$645 + \$90 = \$735.

D) [WRONG] \$745 = \$645 + \$100. Uses incorrect additional charge of \$100.

TRAP ALERT:

Adding the \$150 setup fee again for the revision hours. The setup fee was already included in the \$645 quote.

KEY INSIGHT:

When adding to an existing quote, only add the MARGINAL cost (additional hours x rate). Fixed components don't change.

FAST METHOD:

Two extra hours x \$45/hr = \$90 more. \$645 + \$90 = \$735.

Q48. Correct Answer: C HARD

STEP-BY-STEP SOLUTION:

- At q=0 (launch): S=3.2(0)-1.6=-1.6 thousand = -1,600 units.
- A negative cumulative sales value at launch is not physically meaningful.
- It is a mathematical artifact of fitting a linear model to later data.
- The -1.6 is the y-intercept: the model's predicted value when q=0.

WHY EACH OPTION IS WRONG:

A) [WRONG] Partially correct (not meaningful) but frames it as a factual statement rather than explaining it as a y-intercept extrapolation artifact.

B) [WRONG] The slope is +3.2 (sales INCREASE each quarter). -1.6 is the y-intercept, not the slope.

C) [CORRECT] CORRECT. The y-intercept -1.6 (thousands) means the model predicts -1,600 units at q=0: a modeling limitation.

D) [WRONG] -1.6 as a break-even requirement per quarter is not what a y-intercept represents.

TRAP ALERT:

Confusing slope with y-intercept. The -1.6 is constant (y-intercept), not the per-quarter rate of change.

KEY INSIGHT:

In y=mx+b, b is the predicted value when x=0. Negative b = mathematical deficit at x=0, often a modeling artifact.

FAST METHOD:

At q=0: S=-1.6 thousand. This is the y-intercept: the model's predicted value at launch = -1,600 units.

Q49. Correct Answer: A HARD

STEP-BY-STEP SOLUTION:

- Slope (rate per gallon): m = (108 - 63) / (7,800 - 4,200) = 45 / 3,600 = 0.0125 $/gallon.
- Y-intercept: 63 = 0.0125(4,200) + b => 63 = 52.5 + b => b = 10.5.
- Model: B = 0.0125g + 10.50.
- Verify: at g=7,800: B=0.0125(7,800)+10.50=97.50+10.50=$108. Correct.

WHY EACH OPTION IS WRONG:

A) [CORRECT] CORRECT. Slope=$0.0125/gallon; fixed monthly charge=$10.50.

B) [WRONG] B=0.025g-10.50: uses double the rate and wrong sign on intercept.

C) [WRONG] B=0.0125g-10.50: correct rate but WRONG sign on intercept (should be +10.50, not -10.50).

D) [WRONG] B=0.025g+10.50: uses double the rate (0.025 instead of 0.0125).

TRAP ALERT:

Computing rate as 45/1,800=0.025 (using half the gallon difference, e.g., dividing 3,600 by 2 in error).

KEY INSIGHT:

When finding slope from two data points, always use FULL differences: delta(B)=45, delta(g)=7,800-4,200=3,600.

FAST METHOD:

m=45/3,600=0.0125. b=63-0.0125(4,200)=63-52.5=10.5. B=0.0125g+10.5.

Q50. Correct Answer: B — HARD

STEP-BY-STEP SOLUTION:

- Set D2 >= D1: 2.6y + 14.8 >= 1.8y + 22.4.
- 0.8y >= 7.6 => y >= 9.5.
- First integer: y = 10.
- Year: 2005 + 10 = 2015.
- Verify: y=9: D1=1.8(9)+22.4=38.6M, D2=2.6(9)+14.8=38.2M (D1 ahead). y=10: D1=40.4M, D2=40.8M (D2 ahead).

WHY EACH OPTION IS WRONG:

A) [WRONG] 2014 (y=9): D2=$38.2M < D1=$38.6M. D2 still behind.

B) [CORRECT] CORRECT. 2015 (y=10): D2=$40.8M > D1=$40.4M. First year D2 equals or exceeds D1.

C) [WRONG] 2016 works but is NOT the first year D2 >= D1.

D) [WRONG] 2017 also works but is later.

TRAP ALERT:

Rounding y=9.5 DOWN to 9 and finding D2<D1 at y=9. Must round UP to y=10.

KEY INSIGHT:

At exact crossover y=9.5, budgets are equal. First INTEGER beyond this (y=10) is the first year D2>=D1.

FAST METHOD:

0.8y>=7.6 => y>=9.5 => y=10 => year=2005+10=2015.

TOPIC SUMMARY — CREATING LINEAR MODELS FROM WORD PROBLEMS

COMMON MISTAKES (Top 3):

1. Swapping the flat fee (y-intercept) and per-unit rate (slope). Ask: 'Is this paid ONCE or EVERY unit?' Once = b. Every unit = m.

2. Using the wrong sign. Decreasing quantities (burning, draining, selling, depreciating) require a NEGATIVE slope. 'Burns down' = negative rate.

3. Rounding in the wrong direction for inequality problems. For 'minimum whole number,' always round UP (ceiling). Verify by checking both sides of the crossover.

MUST REMEMBER (Top 3 rules):

1. Linear model structure: y = (rate)(variable) + (starting value). Rate = slope = m (paid per unit). Starting value = y-intercept = b (paid once or initial amount).

2. Break-even/equal-cost: Set the two expressions equal and solve. Net profit = Revenue minus ALL costs (both fixed overhead and variable per-unit costs).

3. Interpreting in context: Slope = 'change in output per 1-unit change in input.' Y-intercept = 'value when x = 0 (initial or one-time amount).' Negative y-intercept = model artifact when extrapolated before the data range.

EXPECTED ON DIGITAL SAT:

3 to 5 questions per test on linear models from word problems (across both Math modules).

Common contexts: pricing and cost models, population or salary growth, physical rates (speed, temperature change, fluid drain), savings and earnings comparisons, depreciation.

Mix of 4-choice MCQ and Grid-In. Higher difficulty versions combine two linear models (break-even, crossover timing, net rate problems, delayed-start comparisons).

■■ END OF SOLUTIONS DOCUMENT ■■